Design and Implementation of a Bluetooth-Controlled Robotic Soccer System

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Abstract

Robot soccer is a challenging application of mobile robotics because it combines mechanical design, embedded systems, wireless communication, motor control, sensing, and decision-making. The objective of this project is to design and develop a Bluetooth-controlled Robo Soccer robot capable of moving in different directions, controlling a lightweight soccer ball, and attempting to score goals on a miniature soccer field. The proposed system uses an Arduino Uno microcontroller as the main controller and an HC-05 Bluetooth module for wireless communication with an Android smartphone. The robot is driven using DC geared motors through an L298N motor driver. A front-mounted dribbling mechanism allows the robot to control the movement of the ball during gameplay. The robot is controlled using a Bluetooth application that enables forward, backward, left, right, and stop movements. The project demonstrates the practical implementation of embedded systems, motor control, and wireless communication in a robotic application. The developed Robo Soccer robot provides an economical platform for learning robotics, embedded programming, motor control, wireless communication, and robotic competitions. The project can be further enhanced by integrating computer vision, artificial intelligence, and autonomous navigation algorithms.

Keywords: Robo Soccer, Arduino UNO, HC-05 Module.

1.Introduction

Robotics has become an important field of engineering because of its applications in industrial automation, healthcare, agriculture, defence, logistics, and other areas. Among various robotic applications, Robo Soccer provides a challenging platform that combines robotics, sensing, control, artificial intelligence, and multi-agent systems [1], [2]. Robot soccer competitions challenge robots to perform tasks such as ball control, navigation, decision-making, cooperation, and goal scoring. RoboCup was introduced as a challenge problem for artificial intelligence and intelligent robotics, providing a common platform for evaluating different technologies and algorithms [1].

Robo Soccer has subsequently become an important research area for investigating autonomous agents, multi-agent cooperation, real-time decision-making, and robotic control

[2], [7]. In this project, a Bluetooth-controlled Robo Soccer robot has been designed and implemented using an Arduino Uno microcontroller. Unlike fully autonomous soccer robots that require computer vision and sophisticated decision-making algorithms, the proposed robot is manually controlled through Bluetooth communication using an Android smartphone.

The robot consists of a four-wheel-drive mechanism powered by DC geared motors. The motors are controlled using an L298N motor driver, while the HC-05 Bluetooth module receives movement commands from a mobile application. A front-mounted dribbling mechanism helps the robot maintain control of the ball and direct it toward the opponent's goal.

1.1 The primary objectives of this project are:

- To design and construct a compact, Arduino-based soccer robot with an integrated dribbling mechanism for enhanced ball control.
- To establish reliable, low-latency teleoperation using an HC-05 Bluetooth module for real-time field versatile.
- To validate the practical integration of embedded control, actuation, and wireless communication in a competitive robotics environment.

The developed system is economical, easy to construct, and suitable for engineering students interested in robotics and embedded systems. It serves as a foundation for future research in autonomous robot soccer, where Bluetooth control can be replaced with computer vision, artificial intelligence, and real-time path planning techniques.

2. Literature Review

Robot soccer has emerged as an important research area in robotics and artificial intelligence because it provides a challenging environment in which different technologies such as robotic control, sensing, decision-making, navigation, multi-agent cooperation, and real-time computation can be integrated into a single application. Unlike conventional mobile robots that operate in relatively predictable environments, soccer robots must operate in a dynamic environment containing a moving ball, other robots, changing positions, and time-dependent decisions. RoboCup was introduced to provide a standard research problem through which artificial intelligence and robotics technologies could be developed, integrated, and evaluated [1].

Kitano et al.[1] introduced RoboCup as a challenge problem for artificial intelligence and intelligent robotics. The authors proposed robot soccer as a common platform for investigating several important areas of AI, including autonomous-agent design, multi-agent collaboration, strategy acquisition, real-time reasoning, robotics, and sensor fusion [1]. The work established robot soccer not only as a competition but also as a research environment in which different robotic technologies could be evaluated under a common problem. This concept has contributed significantly to the development of intelligent robotic systems.

The development of RoboCup subsequently encouraged research into autonomous robot behaviour, coordination, perception, navigation, and strategy. Robot soccer requires a robot to determine its position, understand the state of the game, control its movement, interact with the ball, and respond to other robots. Therefore, the overall control architecture of a soccer robot generally involves several interconnected functions, including perception, decision-making, path planning, obstacle avoidance, and motion control [2], [3].

Abiyev et al.[2] investigated robot soccer control using behaviour trees and fuzzy logic [2]. Their work focused on the decision-making problem of soccer robots operating in a rapidly changing environment. The authors proposed the integration of behaviour trees with fuzzy-logic-based obstacle avoidance. Behaviour

trees were used to organize robot behaviours and decision-making, while fuzzy logic was applied to obstacle avoidance and navigation. The proposed algorithms were evaluated through simulation and real robot implementation. The research demonstrated that combining behaviour-based decision making with fuzzy control can provide flexible responses in dynamic soccer environments [2].

The work of Abiyev et al. is particularly relevant to Robo Soccer because it demonstrates the complexity involved in developing fully autonomous soccer robots. Their system involved perception, decision-making, path finding, obstacle avoidance, and motion control. Such an architecture provides advanced autonomous operation, but it also increases the complexity of both hardware and software. In comparison, the present project uses a simpler control architecture in which the human operator makes the major gameplay decisions through a Bluetooth-connected Android application.Another important research direction in robot soccer is reinforcement learning. Reinforcement learning enables an agent to learn suitable actions through interaction with an environment instead of depending entirely on manually programmed rules.

Martins et al.[3] developed the rSoccer framework for studying reinforcement learning in small and very-small-size robot soccer [3]. The framework provides a simulation environment for evaluating reinforcement-learning algorithms in robot soccer and supports the study of both single-agent and multi-agent behaviours.

The rSoccer framework addresses an important problem in robotic reinforcement learning. Training reinforcement-learning agents directly on physical robots can require a large number of interactions, which may result in increased hardware wear, energy consumption, and development time. Simulation provides an alternative environment in which learning algorithms can be trained and evaluated before being transferred to real robotic systems [3]. The authors demonstrated the use of reinforcement-learning algorithms for learning robot soccer skills and discussed the limitations of these approaches in different scenarios.

Bassani et al.[4] further investigated reinforcement learning and simulation-to-real transfer in robot soccer through the VSSS-RL framework [4]. Their work focused on the IEEE Very Small Size Soccer environment and introduced a simulated environment in which continuous or discrete control policies could be trained. The researchers also proposed a sim-to-real approach for transferring learned policies from simulation to physical robots [4]. Their results demonstrated that reinforcement-learning methods could produce behaviours that would be difficult to implement using only manually designed control policies.

The sim-to-real problem is particularly important in robotics because a simulation cannot perfectly reproduce the physical characteristics of a real robot. Differences in friction, motor behaviour, mechanical backlash, sensor noise, actuator response, communication delay, and other physical factors can create a gap between simulation and reality [4]. Therefore, although simulation-based reinforcement learning can reduce the time and cost associated with physical training, additional techniques are required before learned policies can reliably operate on real robots.

Recent research has also investigated deep reinforcement learning combined with visual perception for robot soccer. Vision-based approaches allow robots to obtain information about the ball, other robots, and the surrounding environment using cameras. Deep reinforcement learning can then be used to learn complex behaviours and decision-making strategies from these observations. Such approaches represent a major step toward highly autonomous robot soccer systems, but they generally require more computational resources, sophisticated software, and suitable

sensing hardware than basic educational robot platforms.

The development of autonomous soccer robots therefore involves a trade-off between system intelligence and implementation complexity. Advanced systems based on computer vision, reinforcement learning, and autonomous decision-making can provide sophisticated gameplay capabilities, but they require additional hardware and software development. In contrast, manually controlled robotic platforms can provide a simpler method for demonstrating the fundamental concepts of robot movement, motor control, wireless communication, and embedded programming.

The literature indicates that robot soccer research has progressively moved from basic robotic control toward increasingly intelligent autonomous systems. Early RoboCup research emphasized the integration of robotics and artificial intelligence [1]. Later work investigated behaviour-based decision-making and fuzzy obstacle avoidance [2]. More recent research has explored reinforcement learning, simulation environments, and sim-to-real transfer [3], [4]. These developments demonstrate the increasing complexity and intelligence of modern Robo Soccer systems.

However, most advanced research systems focus on autonomous operation and require complex algorithms, sensing systems, and computational platforms. For undergraduate engineering education and low-cost robotics competitions, a simpler platform can be useful because it allows students to understand the fundamental relationship between wireless communication, embedded controllers, motor drivers, actuators, and mechanical mechanisms before progressing to autonomous robotics.

The present project therefore adopts a different approach by developing a low-cost Bluetooth-controlled Robo Soccer robot. Instead of using complex computer-vision or reinforcement-learning algorithms for real-time decision-making, the proposed system uses an Android smartphone as the user interface. Movement commands are transmitted wirelessly through an HC-05 Bluetooth module to an Arduino Uno microcontroller. The Arduino processes the received commands and controls the DC geared motors through an L298N motor driver.

This approach reduces the computational complexity of the robot while retaining the important fundamental features of a mobile robotic system. The operator can control forward, backward, left, right, and stop movements, while the mechanical dribbling mechanism assists in controlling the soccer ball. The system therefore provides a practical platform for demonstrating embedded programming, wireless communication, motor control, and mobile robot operation.

4.Research Gap

Based on the reviewed literature [3]–[6], the following research gaps were identified:

• Many Robo Soccer systems focus on autonomous control, artificial intelligence, computer vision, and machine learning.

• Advanced autonomous systems require more sophisticated hardware and software.

• Simulation and reinforcement-learning frameworks are useful for advanced research but require additional machine-learning knowledge [4], [5].

• Vision-based deep reinforcement learning requires onboard sensing and computational resources [6].

• There is an educational need for simpler robotic platforms that demonstrate fundamental concepts such as wireless communication, motor control, and embedded programming.

The proposed project addresses this educational requirement by developing a compact Bluetooth-controlled Robo Soccer robot using an Arduino Uno, HC-05 Bluetooth module, L298N motor driver, and DC geared motors.

5.Problem Statement

Robot soccer systems that use artificial intelligence, computer vision, and autonomous navigation can provide advanced functionality but may require sophisticated hardware and software. The complexity of such systems can make them difficult to implement in undergraduate engineering projects.

Therefore, there is a need for a compact, low-cost, and easy-to-build Robo Soccer robot that demonstrates the fundamental concepts of robotics, embedded systems, wireless communication, and motor control without requiring advanced image-processing techniques.

The proposed Bluetooth-controlled Robo Soccer robot aims to provide an economical solution in which the user controls the robot using an Android smartphone while maintaining smooth movement and effective ball handling.

6.Objectives of the Proposed Work

The main objectives of this project ares

1. To design and fabricate a compact Bluetooth-controlled Robo Soccer robot.
2. To implement wireless communication using the HC-05 Bluetooth module.
3. To control four DC geared motors using the Arduino Uno and L298N motor driver.
4. To develop a simple front-mounted dribbling mechanism for controlling the soccer ball.
5. To evaluate the robot's movement and Bluetooth communication through experimental testing.
6. To provide an affordable robotics platform suitable for engineering education and student competitions.
7. To create a prototype that can be upgraded in the future with computer vision, artificial intelligence, and autonomous navigation.

Author	Year	Research Focus	Key Finding	Limitation
Kitano et al. [1]	1997	RoboCup and AI	Established robot soccer as an AI research challenge	Focuses on autonomous/intelligent systems
Veloso et al. [2]	2000	RoboCup research	Highlighted RoboCup as a platform for robotics research	Advanced research enviroment
Abiyev et al. [3]	2016	Behaviour trees and fuzzy logic	Improved robot-soccer decision making and obstacle avoidance	Requires autonomous control algorithms
Martins et al. [4]	2021	Reinforcement learning	Developed rSoccer framework for robot soccer	Requires reinforcement

				-learning knowledge
Bassani et al. [5]	2020	Sim-to-real learning	Demonstrated simulation-to-real transfer	Requires simulation and ML techniques
Tirumala et al. [6]	2024/2025	Deep RL and vision	Demonstrated end-to-end vision-based robot soccer	Computationally demanding

7.Proposed System Architecture

The proposed Robo Soccer system is designed as a Bluetooth-controlled mobile robot capable of maneuvering on a miniature soccer field and controlling a lightweight ball. The system integrates embedded control, wireless communication, motor control, and mechanical ball handling.

The Arduino Uno acts as the central controller. Movement commands are transmitted from an Android smartphone to the HC-05 Bluetooth module. The Arduino processes the received command and generates control signals for the L298N motor driver. The motor driver controls the DC geared motors according to the selected movement command.

Block Diagram:

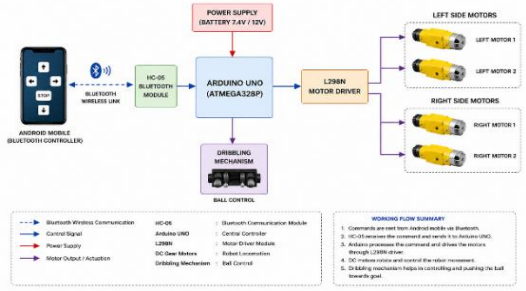


Fig 7.1.Block Diagram of the Bluetooth-Controlled Robo Soccer Robot

8.Hardware Components

8.1.Arduino Uno

The Arduino Uno acts as the brain of the robot. It processes commands received through Bluetooth and controls the motors accordingly. It is based on the ATmega328P microcontroller and provides digital and analog input/output pins for interfacing with sensors and actuators.

Specifications

- Operating Voltage: 5 V
- Microcontroller: ATmega328P
- Digital I/O Pins: 14
- Analog Inputs: 6
- Clock Frequency: 16 MHz

8.2.HC-05 Bluetooth Module:

The HC-05 module enables wireless communication between the robot and an Android smartphone. Commands such as Forward, Backward, Left, Right, and Stop are transmitted wirelessly.

Advantages:

- Low power consumption
- Reliable communication

- Easy Arduino interfacing
- Approximately 10-meter operating range

8.3. L298N Motor Driver

Since Arduino cannot directly drive DC motors, the L298N motor driver is used. It controls the direction and speed of the motors according to signals from the Arduino.

Functions:

- Forward rotation
- Reverse rotation
- Speed control using PWM
- Dual H-Bridge motor driving

8.4. DC Gear Motors

Four DC geared motors provide movement to the robot.

Advantages include:

- High torque
- Stable operation
- Low power consumption
- Good maneuverability

8.5. Power Supply:

The robot is powered using a rechargeable battery pack that supplies sufficient voltage to the Arduino, Bluetooth module, motor driver, and motors.

9.Working Principle

The working of the proposed Robo Soccer robot is illustrated as follows:

1. The smartphone connects to the HC-05 Bluetooth module.
2. The user presses a movement button.
3. The command is transmitted through Bluetooth.
4. The HC-05 receives the command.
5. The command is transferred to the Arduino Uno.
6. Arduino processes the command.
7. Arduino sends control signals to the L298N motor driver.
8. The motor driver operates the DC geared motors.
9. The robot moves in the selected direction.
10. The dribbling mechanism assists in controlling the ball.
11. The operator manually directs the robot toward the goal.

This sequence repeats continuously throughout the game.

10.Software Design

The robot is programmed using the **Arduino IDE** in Embedded C.

The software continuously waits for Bluetooth commands and executes motor control functions.

Algorithm

Step 1: Start

Step 2: Initialize Arduino pins

Step 3: Initialize Bluetooth module

Step 4: Initialize motor driver

Step 5: Read Bluetooth command

Step 6: If command = F

→ Move Forward

Step 7: Else If command = B

→ Move Backward

Step 8: Else If command = L

→ Turn Left

Step 9: Else If command = R

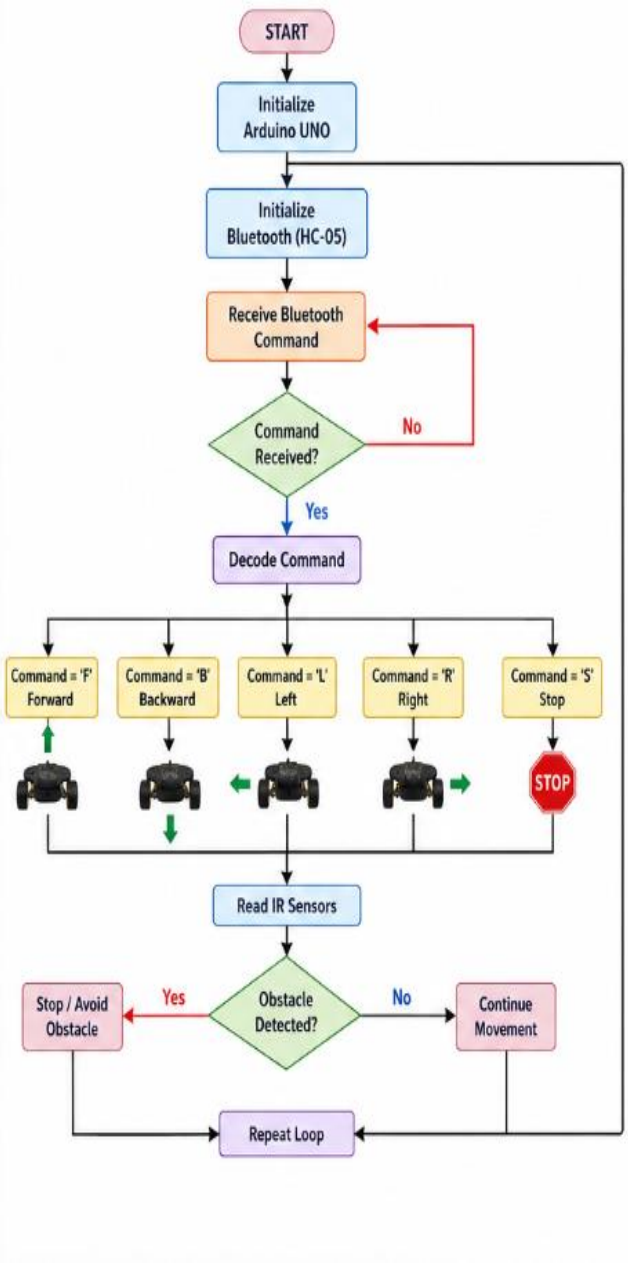
→ Turn Right

Step 10: Else If command = S

→ Stop Robot

Step 11: Read IR Sensors

Step 12: Repeat continuously



11.Experimental Results and Performance Analysis

The proposed Bluetooth-controlled Robo Soccer robot was designed, fabricated, and experimentally tested on a miniature indoor soccer field. The objective of the testing was to evaluate the robot's movement, wireless communication, ball-handling capability, and overall system performance. The robot was controlled using an Android smartphone connected to the HC-05 Bluetooth module.

The experimental setup consisted of an Arduino Uno, HC-05 Bluetooth module, L298N motor driver, four DC geared motors, two IR sensors, a rechargeable battery pack, and a lightweight plastic soccer ball. Multiple trials were conducted to verify the robot's reliability under different operating conditions.

Table-11: Experimental Results-

Parameter	Observed Result
Bluetooth Communication Range	10 meters
Robot Response Time	Less than 1 second
Forward Movement	Successful
Backward Movement	Successful
Left Turn	Successful
Right Turn	Successful
Stop Command	Successful
Ball Dribbling	Stable
Goal Scoring	Successfully demonstrated
Obstacle Detection	Accurate within short range
Battery Backup	Approximately 1.5–2 hours

The experimental results indicate that the proposed robot provides smooth motion, stable wireless communication, and effective ball control during gameplay. The response time between the smartphone and robot was found to be very low, allowing real-time operation.

12.Robot Movement Analysis

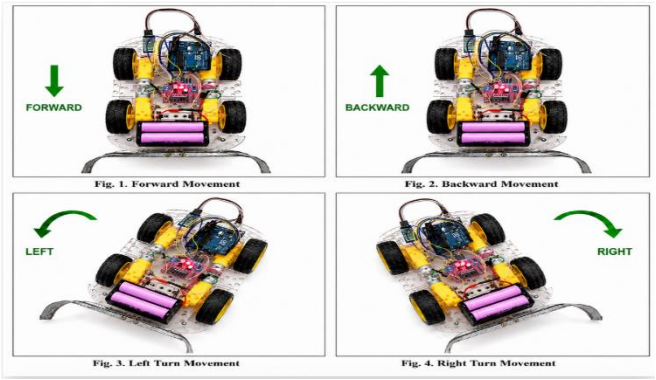


Fig 12.1. Robot Movement Control in Forward, Backward, Left, and Right Directions

13.Performance Analysis

The overall performance of the developed Robo Soccer robot was analyzed based on communication reliability, mobility, control accuracy, and ease of operation.

Communication Performance:

The HC-05 Bluetooth module maintained a stable wireless connection up to approximately 10 meters without noticeable communication delay.

Motor Performance:

The L298N motor driver successfully controlled the direction and speed of the four DC geared motors. The robot demonstrated smooth forward, backward, left, and right movement with minimal vibration.

Ball Handling:

The front-mounted dribbling mechanism effectively maintained possession of the ball during movement. The lightweight structure of the robot allowed better maneuverability around the soccer field.

Overall System Performance:

The developed prototype achieved the project objectives by providing a compact, economical, and reliable Bluetooth-controlled Robo Soccer robot suitable for educational purposes and robotics competitions.

Advantages

The proposed Robo Soccer robot offers several advantages:

- Low-cost and economical design.
- Easy to assemble and maintain.
- Reliable Bluetooth wireless communication.
- Fast response to user commands.
- Simple embedded system architecture.
- Compact and lightweight construction.
- Smooth movement in all directions.
- Effective ball control using the dribbling mechanism.
- Low power consumption.
- Suitable for robotics competitions and engineering education.

Applications

The developed Robo Soccer robot can be applied in several areas:

- Robo Soccer competitions.
- Engineering laboratory experiments.
- Robotics workshops.
- Embedded systems education.
- Wireless communication demonstrations.
- Mobile robotics research.
- Student innovation projects.
- STEM education and training.
- Prototype development for autonomous robots.

14.Future Scope

Although the proposed robot performs effectively under Bluetooth control, several improvements can be incorporated in future versions:

1. Integration of computer vision using OpenCV.
2. Artificial intelligence-based autonomous gameplay.
3. Machine learning algorithms for strategy optimization.
4. Automatic ball detection using cameras.
5. Real-time path planning and obstacle avoidance.
6. GPS or indoor localization techniques.
7. Wi-Fi or ESP32-based communication.
8. Team coordination among multiple robots.
9. Voice-controlled operation.
10. Internet of Things (IoT) monitoring and remote control.

These enhancements will significantly improve the intelligence and autonomy of the Robo Soccer system.

15.Conclusion

This research presented the design and implementation of a **Bluetooth-Controlled Robo Soccer Robot** using an Arduino Uno microcontroller. The proposed system successfully integrates Bluetooth communication, motor control, embedded programming, obstacle detection, and a dribbling mechanism into a compact robotic platform.

Experimental evaluation demonstrated reliable wireless communication, smooth robot movement, effective ball handling, and accurate obstacle detection. The HC-05 Bluetooth module provided stable connectivity with an Android smartphone, while the L298N motor driver ensured efficient operation of the DC geared motors. The robot successfully executed movement commands and demonstrated goal-scoring capability on a miniature soccer field.

Compared with autonomous Robo Soccer systems that require advanced image processing and artificial intelligence, the proposed design offers a simple, affordable, and practical solution suitable for undergraduate engineering students. The project also provides a strong foundation for future research involving computer vision, reinforcement learning, and autonomous navigation.

In conclusion, the developed Robo Soccer robot successfully achieves the project objectives and serves as an effective educational platform for learning robotics, embedded systems, wireless communication, and mobile robot control.

16.References

[1] H. Kitano et al., "RoboCup: A Challenge Problem for AI," *AI Magazine*, vol. 18, no. 1, pp. 73–85, 1997.

[2] R. H. Abiyev, et al., "Robot Soccer Control Using Behaviour Trees and Fuzzy Logic Procedia Computer Science Volume 102, 2016, Pages 477-484

[2] M. Martins et al., "rSoccer: A Framework for Studying Reinforcement Learning in Small and Very Small Size Robot Soccer," *arXiv*, 2021.

[4] M. Martins et al., "A Framework for Studying Reinforcement Learning and Sim-to-Real in Robot Soccer," *arXiv*, 2020.

[5] B. Su et al., "Learning Robot Soccer from Egocentric Vision with Deep Reinforcement Learning," *arXiv*, 2024.

[6] M. Veloso, P. Stone, and K. Han, "The RoboCup Research Initiative," *Robotics and Autonomous Systems*, 2000.

[7] J. Bruce and M. Veloso, "Real-Time Multi-Robot Strategy in Robo Soccer," *IEEE International Conference on Robotics and Automation*, 2002.

[8] J. Zico Kolter and A. Ng, "Policy Search via Reinforcement Learning for Robot Soccer," *Proceedings of ICRA*, 2008.

[9] IEEE Robotics and Automation Society, "Robot Soccer Systems and Applications," 2023.

[10] Additional recent RoboCup and IEEE conference papers on robot soccer, wireless robotics, and embedded control systems.